

CAP Theorem - AP Databases (Availability + Partition Tolerance)

1. What is an AP System?

AP means:

None

A = Availability

P = Partition Tolerance

System guarantees:

- Every request gets response
- Continues operating during network partition

But may sacrifice:

None

Consistency

Some users may receive stale or conflicting data temporarily.

2. Simple Meaning

If cluster splits:

None

Region A disconnected from Region B

Both sides still serve requests.

But values may differ until sync happens later.

3. Why AP Is Chosen

Used when serving users continuously is more important than perfect freshness.

Examples:

Social media feeds

Product catalogs

Likes/comments counts

Messaging presence status

Recommendation systems

Analytics dashboards

4. Example Scenario

Profile picture changed.

During partition:

None

Node A shows new photo

Node B shows old photo

System still responds everywhere.

After recovery:

None

Nodes synchronize

5. Eventual Consistency

Most AP systems use:

None

Eventual Consistency

Meaning if no new updates occur, all replicas converge to same value after some time.

6. Apache Cassandra

Apache Cassandra is a classic AP-oriented system.

Characteristics:

- Peer-to-peer architecture
- No single primary required
- Multi-node writes possible
- Tunable consistency levels
- High uptime across regions

During partition, nodes can continue serving requests.

7. Amazon DynamoDB Clarification

Amazon DynamoDB is inspired by AP-style distributed principles, but supports multiple consistency modes:

- Eventually consistent reads
- Strongly consistent reads (certain scopes)

So calling it “pure AP always” is oversimplified.

8. Facebook / Social App Example

For apps like Meta Platforms social systems:

None

User wants app working now

Seeing 999 likes instead of 1000 for a few seconds is acceptable.

Availability wins.

9. How AP Systems Work

A. Replication Everywhere

Multiple copies of data.

B. Accept Writes on Many Nodes

Requests don't depend on one leader.

C. Conflict Resolution Later

Methods include:

- Last write wins
- Vector clocks
- Merge logic
- Application reconciliation

10. Trade-offs

Advantages

High uptime

Good geo-distribution

Resilient during network issues

Low user-facing downtime

Disadvantages

Stale reads
Write conflicts
Complex reconciliation
Temporary inconsistency

11. AP vs CP

Feature	AP	CP
Respond during partition	Yes	Sometimes no
Latest data guaranteed	No	Yes
Best for	Feeds/catalog	Payments/banking

12. Interview Answer Example

Q: Why choose AP for social media?

Because continuous availability is more valuable than perfect freshness.
Users prefer seeing slightly stale counts rather than service outages.

13. HLD Use Cases

Choose AP for:

None

Feed service

Search suggestions

Session presence

Likes count

Global content cache

Product browse pages

14. Short Revision Notes

None

AP =

Availability + Partition tolerance

May return stale data

Later sync replicas

Used for:

social apps

catalogs

feeds

analytics

15. One-Line Summary

AP databases prioritize continuous service during network failures, accepting temporary inconsistency and later synchronizing data across nodes.